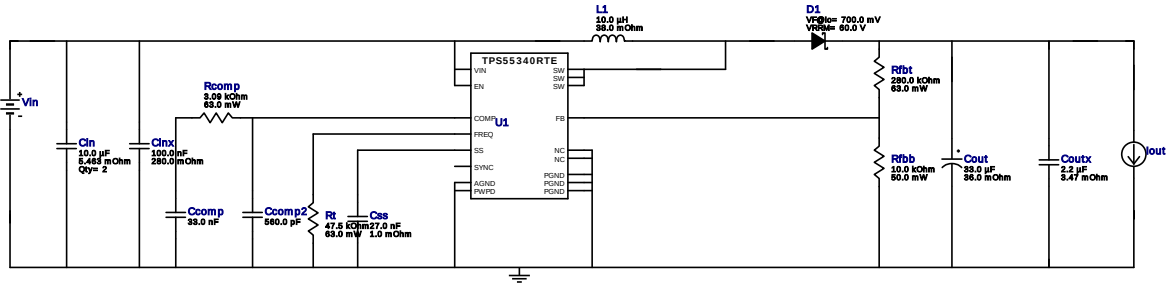


WEBENCH® Design Report

Design : 28 TPS55340RTER
TPS55340RTER 11V-12V to 36.00V @ 0.7A

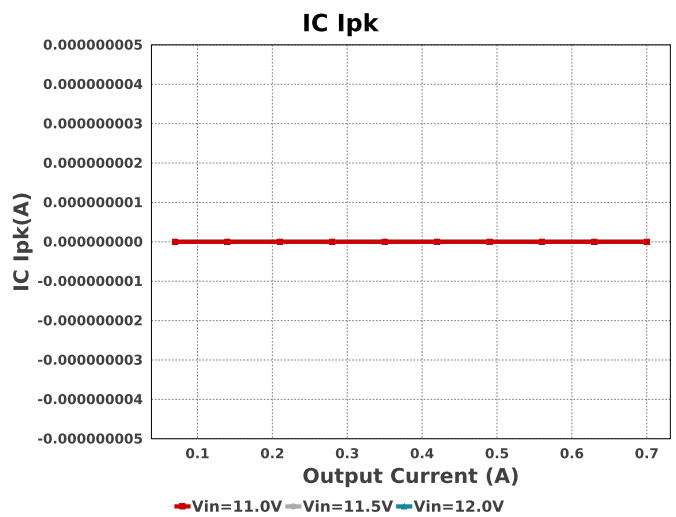
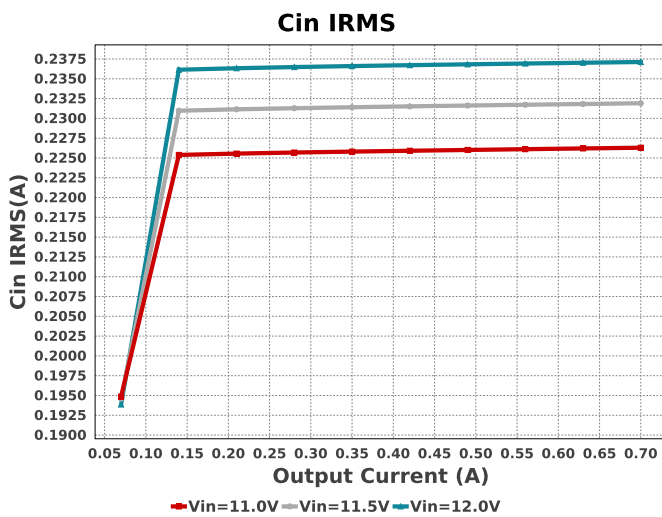
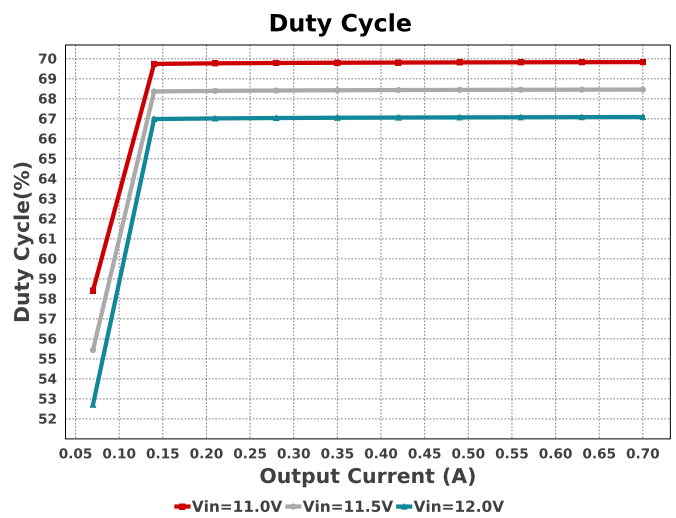
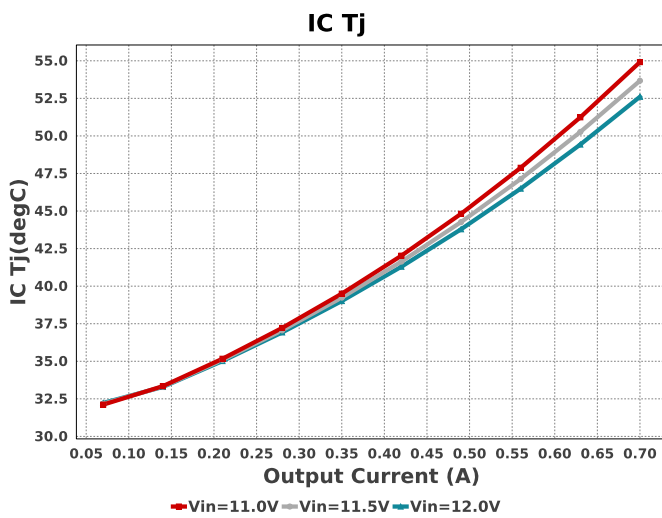


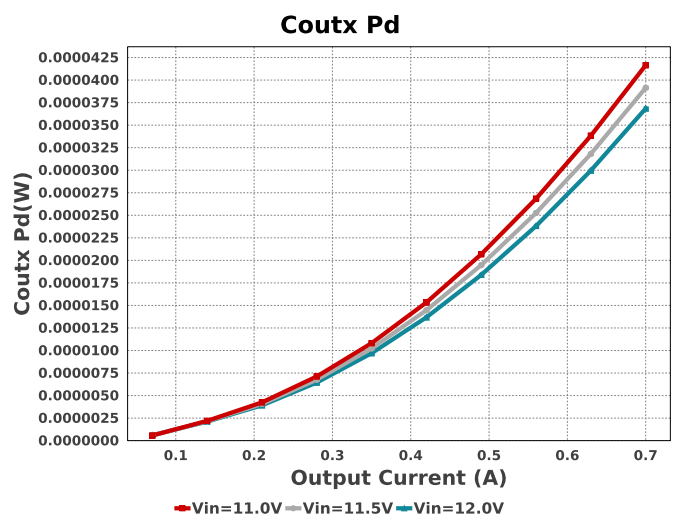
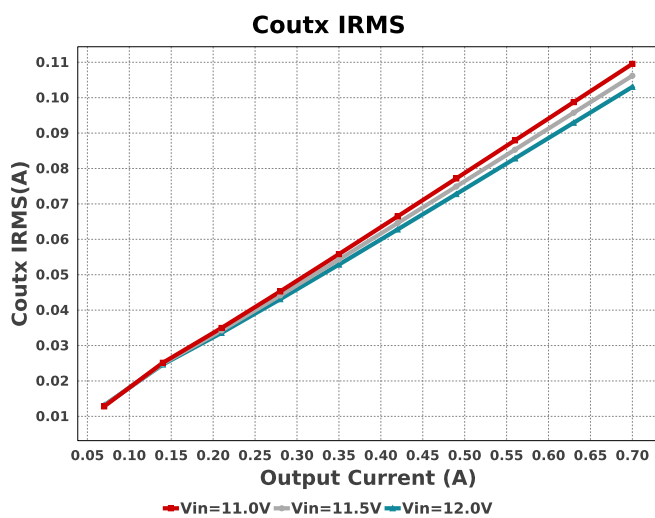
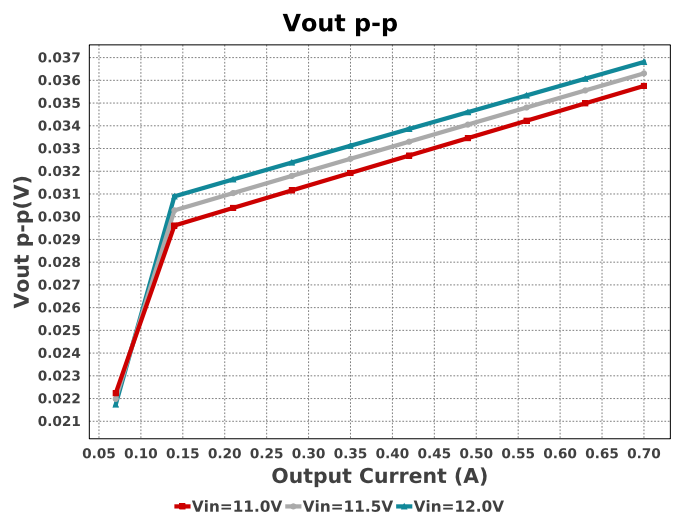
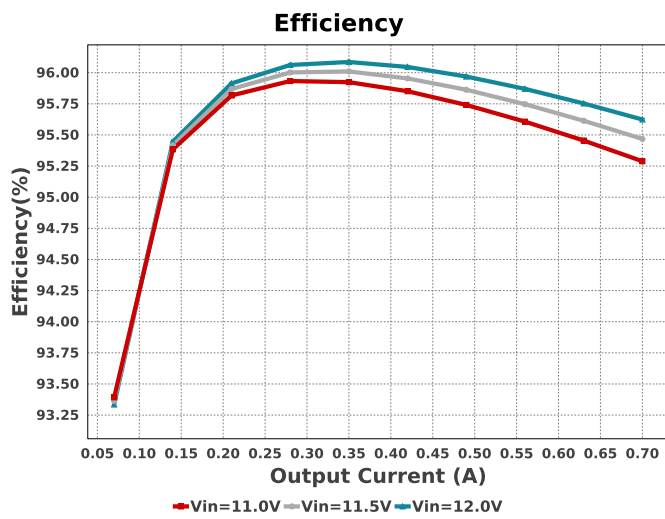
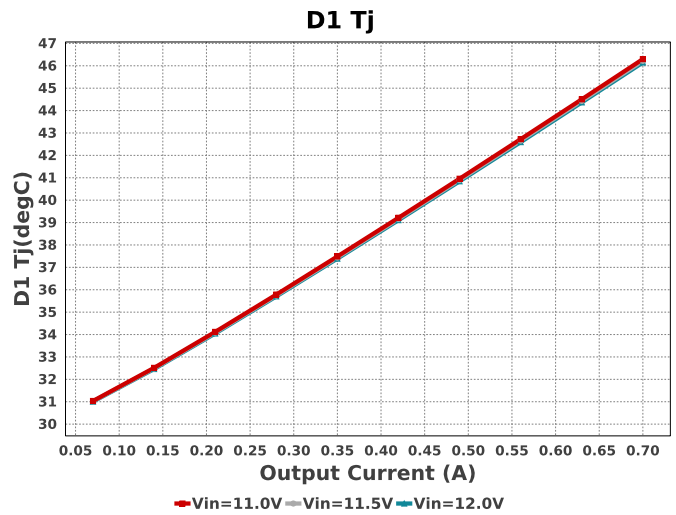
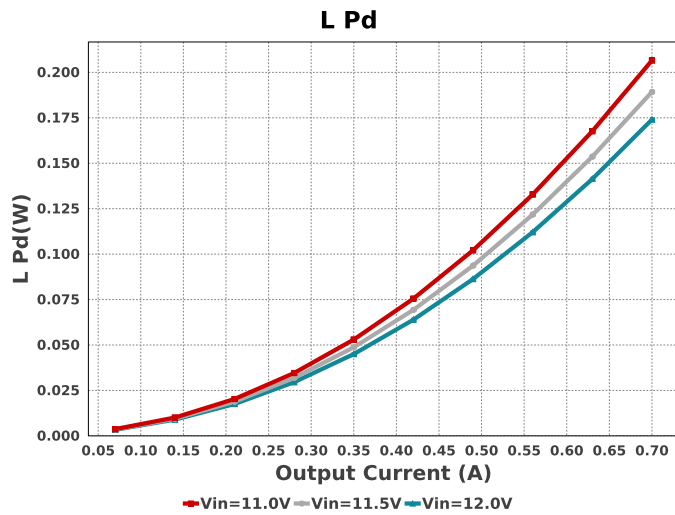
1. For frequency > 1.2 MHz, the user is required to make sure load current is always greater than "Iout Min" value shown in the Op Vals.

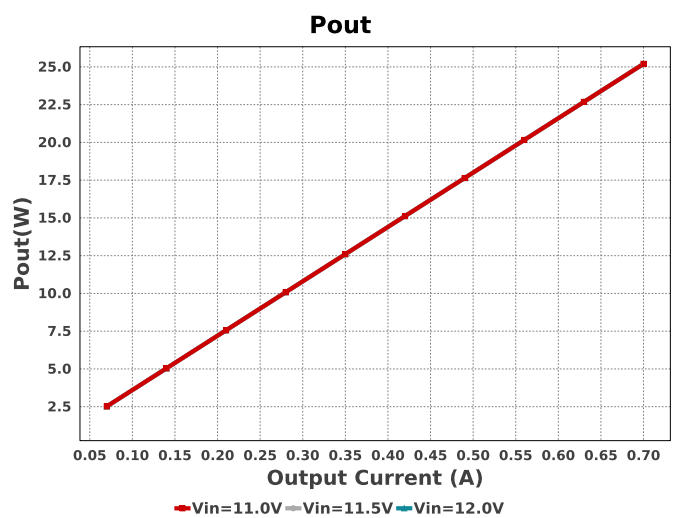
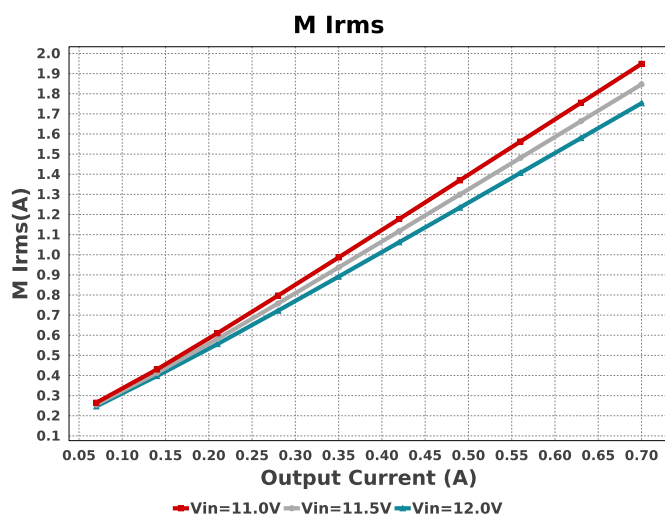
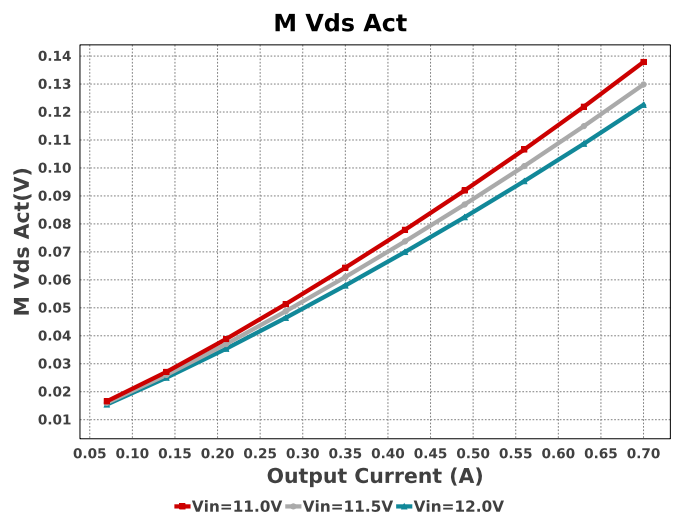
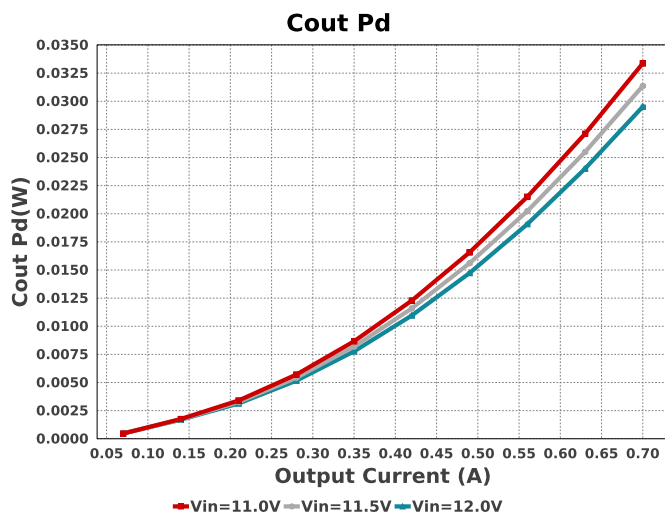
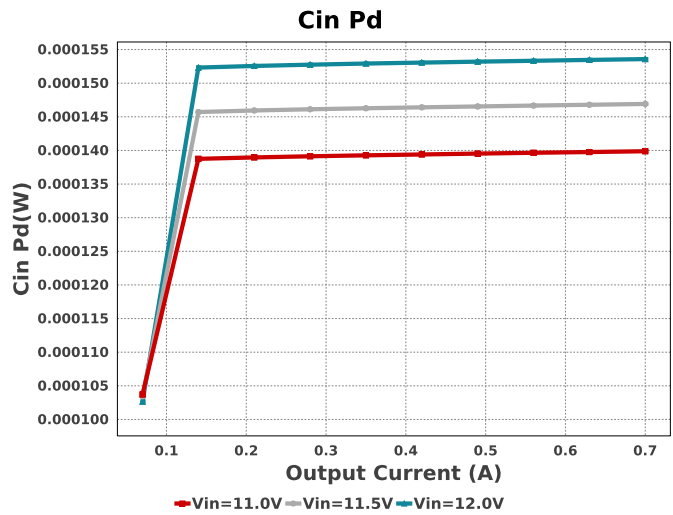
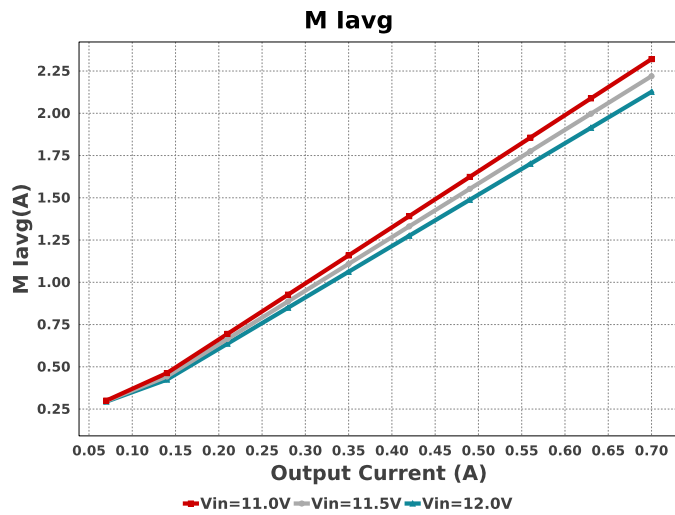
Electrical BOM

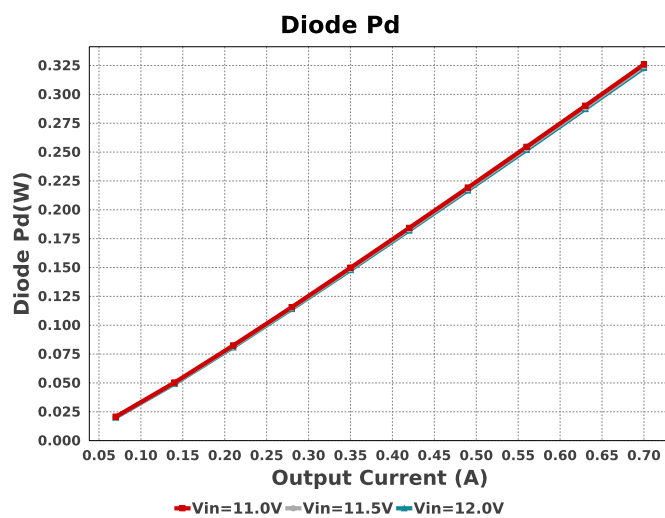
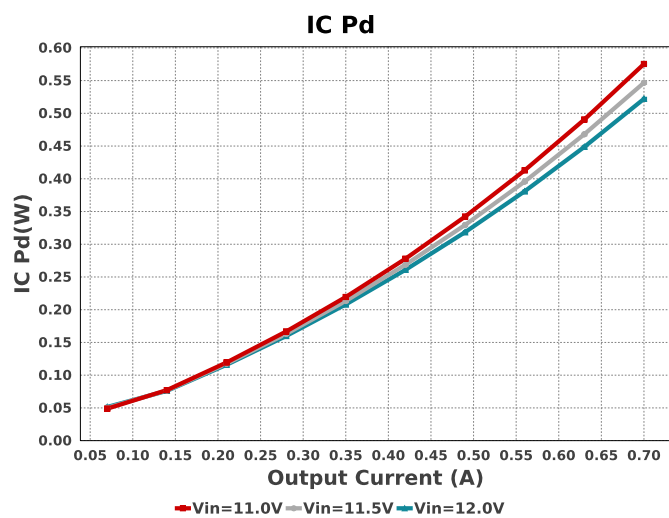
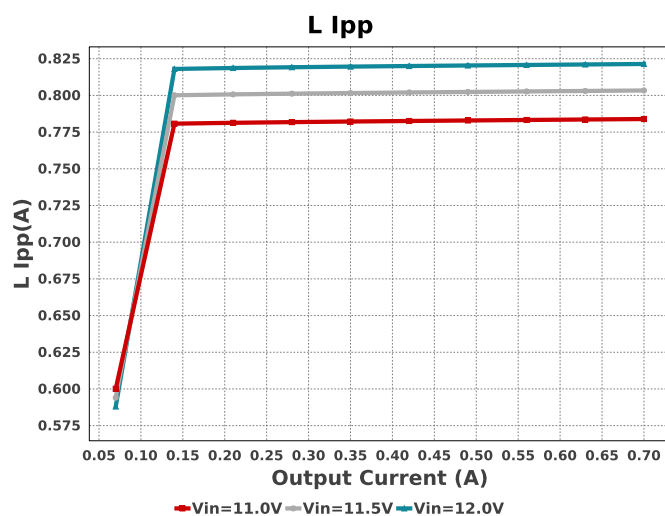
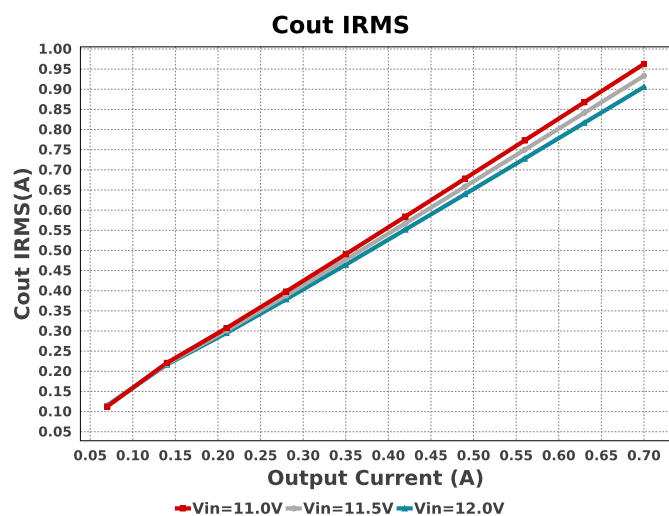
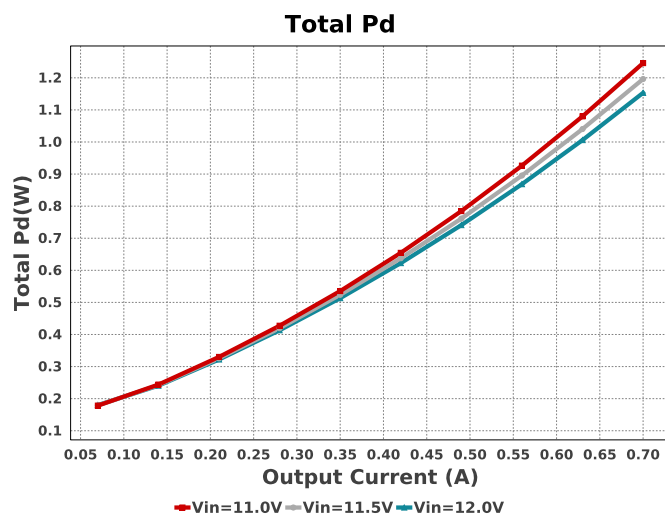
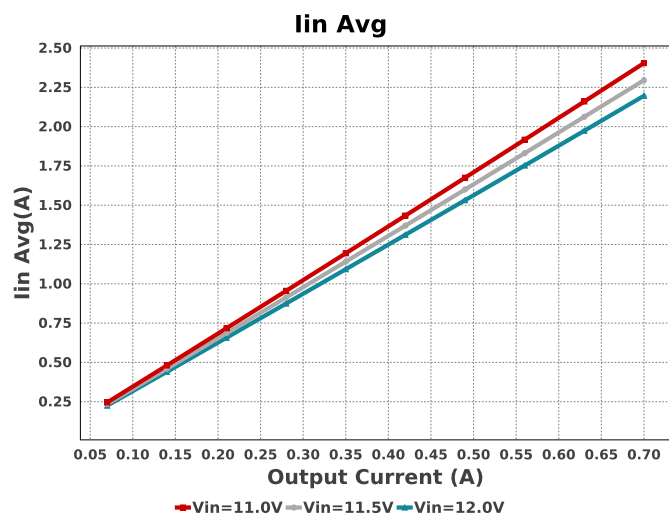
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Ccomp	TDK	CGA4J2C0G1H333J125AA Series= C0G/NP0	Cap= 33.0 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.08	0805 7 mm ²
Ccomp2	MuRata	GRM1555C1H561JA01J Series= C0G/NP0	Cap= 560.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
Cin	Taiyo Yuden	MSAST21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.463 mOhm VDC= 25.0 V IRMS= 2.36768 A	2	\$0.04	0805 7 mm ²
Cinx	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cout	Panasonic	EEHZA1K330P Series= ZA	Cap= 33.0 uF ESR= 36.0 mOhm VDC= 80.0 V IRMS= 1.7 A	1	\$1.34	SM_RADIAL_10BMM 160 mm ²
Coutx	TDK	C2012X5R1H225K125AB Series= X5R	Cap= 2.2 uF ESR= 3.47 mOhm VDC= 50.0 V IRMS= 0.0 A	1	\$0.13	0805 7 mm ²
Css	MuRata	GRM155R71A273KA01D Series= X7R	Cap= 27.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
D1	Diodes Inc.	B560C-13-F	VF@Io= 700.0 mV VRRM= 60.0 V	1	\$0.19	SMC 83 mm ²
L1	NIC Components	NPI31W100MTRF	L= 10.0 uH 38.0 mOhm	1	\$0.36	IND_NPI31W 172 mm ²

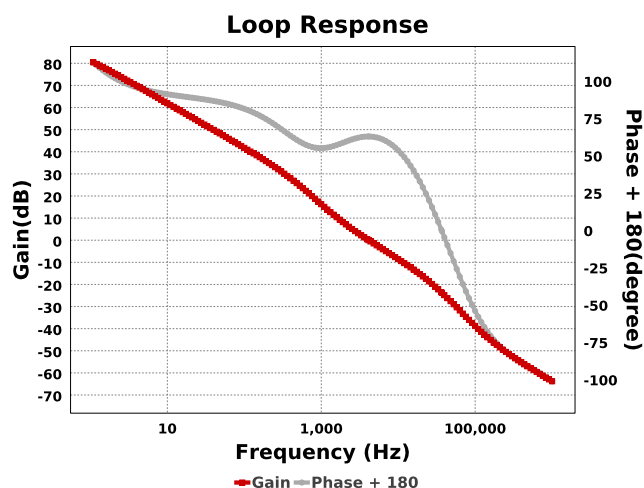
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rcomp	Vishay-Dale	CRCW04023K09FKED Series= CRCW..e3	Res= 3.09 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbb	Yageo	RC0201FR-0710KL Series= ?	Res= 10.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	\$0.01	0201 2 mm ²
Rfbt	Vishay-Dale	CRCW0402280KFKED Series= CRCW..e3	Res= 280.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rt	Vishay-Dale	CRCW040247K5FKED Series= CRCW..e3	Res= 47.5 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
U1	Texas Instruments	TPS55340RTER	Switcher	1	\$1.17	RTE0016C 16 mm ²











Operating Values

#	Name	Value	Category	Description
1.	BOM Count	15		Total Design BOM count
2.	Total BOM	\$3.424		Total BOM Cost
3.	Cin IRMS	226.294 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	139.88 μ W	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	962.748 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	33.368 mW	Capacitor	Output capacitor power dissipation
7.	Coutx IRMS	109.552 mA	Capacitor	Output capacitor_x RMS ripple current
8.	Coutx Pd	41.645 μ W	Capacitor	Output capacitor_x power loss
9.	D1 Tj	46.299 degC	Diode	D1 junction temperature
10.	Diode Pd	325.97 mW	Diode	Diode power dissipation
11.	IC Ipk	0.0 A	IC	Peak switch current in IC
12.	IC Pd	575.37 mW	IC	IC power dissipation
13.	IC Tj	54.913 degC	IC	IC junction temperature
14.	IC Tolerance	9.0 mV	IC	IC Feedback Tolerance
15.	ICThetaJA	43.3 degC/W	IC	IC junction-to-ambient thermal resistance
16.	Iin Avg	2.404 A	IC	Average input current
17.	L Ipp	783.907 mA	Inductor	Peak-to-peak inductor ripple current
18.	L Pd	206.57 mW	Inductor	Inductor power dissipation
19.	M Iavg	2.321 A	Mosfet	MOSFET Average current
20.	M Irms	1.948 A	Mosfet	MOSFET RMS ripple current
21.	M Vds Act	137.886 mV	Mosfet	Voltage drop across the MosFET
22.	Cin Pd	139.88 μ W	Power	Input capacitor power dissipation
23.	Cout Pd	33.368 mW	Power	Output capacitor power dissipation
24.	Coutx Pd	41.645 μ W	Power	Output capacitor_x power loss
25.	Diode Pd	325.97 mW	Power	Diode power dissipation
26.	IC Pd	575.37 mW	Power	IC power dissipation
27.	L Pd	206.57 mW	Power	Inductor power dissipation
28.	Total Pd	1.246 W	Power	Total Power Dissipation
29.	Cross Freq	3.825 kHz	System	Bode plot crossover frequency
30.	Duty Cycle	69.835 %	Information System	Duty cycle
31.	Efficiency	95.289 %	Information System	Steady state efficiency
32.	FootPrint	482.0 mm ²	Information System	Total Foot Print Area of BOM components
33.	Frequency	983.333 kHz	Information System	Switching frequency
34.	Gain Marg	-23.343 dB	Information System	Bode Plot Gain Margin
35.	Iout	700.0 mA	Information System	Iout operating point
36.	Low Freq Gain	80.237 dB	Information System	Gain at 1Hz
37.	Mode	CCM	Information System	Conduction Mode
38.	Phase Marg	62.079 deg	Information System	Bode Plot Phase Margin
39.	Pout	25.2 W	Information System	Total output power
40.	Vin	11.0 V	Information System	Vin operating point

#	Name	Value	Category	Description
41.	Vout	36.0 V	System Information	Operational Output Voltage
42.	Vout Actual	35.641 V	System Information	Vout Actual calculated based on selected voltage divider resistors
43.	Vout Tolerance	2.697 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
44.	Vout p-p	35.753 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	700.0 m	Maximum Output Current
VinMax	12.0	Maximum input voltage
VinMin	11.0	Minimum input voltage
Vout	36.0	Output Voltage
base_pn	TPS55340	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

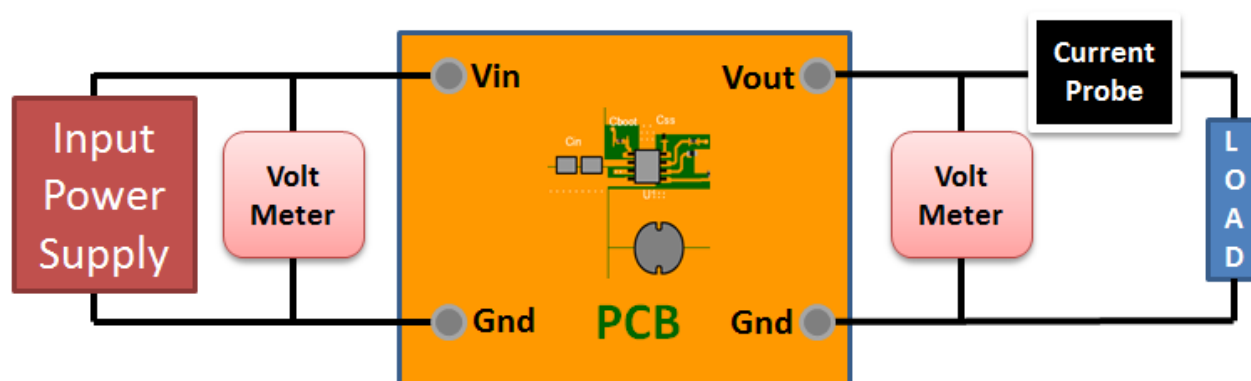
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 11.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

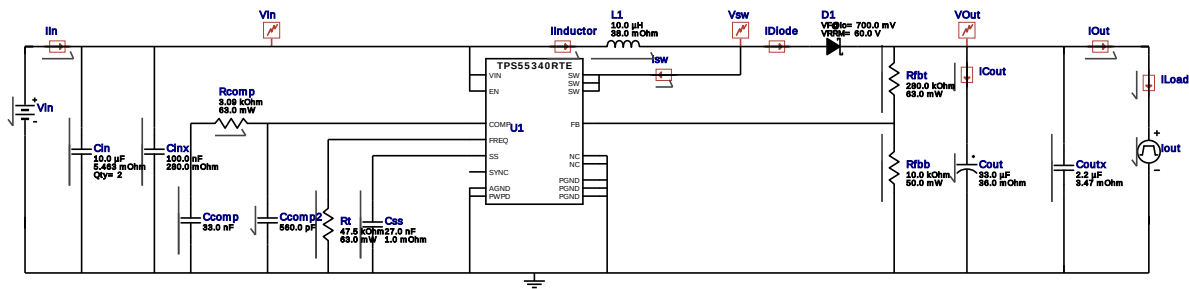
Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



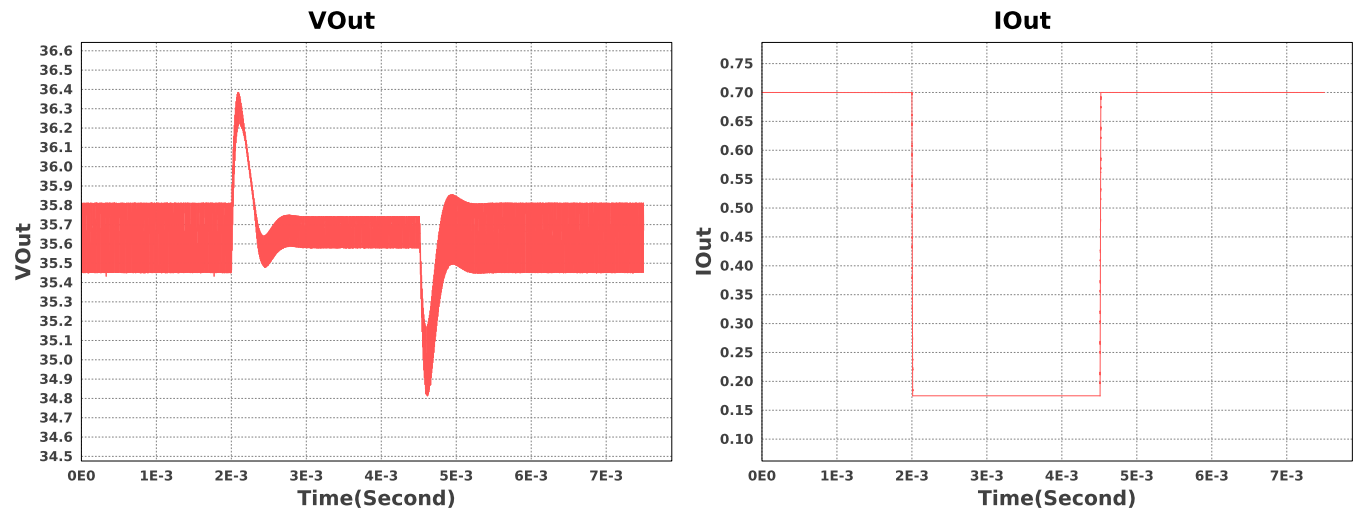
WEBENCH® Electrical Simulation Report

Design Id = 28
sim_id = 8
Simulation Type = Load Transient



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Cout	IC	no description	36.0 V
2.	Iout	signal_type	Signal Type	PULSE
		I1	Initial Current	0.7 A
		I2	Peak Current	0.175 A
		Td	Initial Time Delay	2m Sec
		Tr	Rise Time	10u Sec
		Tf	Fall Time	10u Sec
		Pw	Pulse Width	2.5m Sec



Design Assistance

- Master key : 3DA890648EAF22B9[v1]
- TPS55340 Product Folder : <http://www.ti.com/product/TPS55340> : contains the data sheet and other resources.

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